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METHODOLOGIES AND APPLICATION

psda: A tool for extracting knowledge from symbolic data with an application in Brazilian educational data

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Abstract

Symbolic polygonal data analysis is a new type of framework to extract valuable knowledge from a new structure of data using regular polygon built from data in class, big data, and complex data. This paper introduces a toolbox for symbolic polygonal data, named *psda*, that contains the main descriptive measures for this type of variable, e.g., mean, variance, correlation, and a polygonal linear regression model (plr). It is applied at the Brazilian Basic Education Assessment System (SAEB), giving a new perspective to the managers of the counties to realize the public policy in the Brazilian educational system. The hypothesis test showed that the polygonal linear regression model presented the best performance compared to some symbolic interval regression models in the SAEB application.

Keywords Polygonal data · psda · Symbolic data analysis · Regression · Descriptive measures · R

1 Introduction

Data analysis is a fundamental framework for extraction of knowledge on biology, statistics, computing science, data mining and so on. This statistical approach is composed of many techniques, e.g., mean, variance, correlation, graphics and others developed over the years. For centuries the object of study of data analysis has been a p-dimensional point in $\mathbb{R}^p$; this framework is known as classical data analysis.

From technological advances, the structure of data has been improved every day. Unfortunately, classical data is limited to the study in p-dimensional point. Billard and Diday (2003, 2007) introduces a new type of data considering complex and diverse structures of data, e.g., histogram, probability distributions, intervals, list of categories, etc. This type of data is called symbolic data, and its study is known as symbolic data analysis (SDA). First step in SDA is to build the symbolic dataset, where the rows are subsets of individual entities having a common property, called classes.

In order to take the variability of the individuals inside each class, these new units are described by variables that can take symbolic values. According to Diday (2016) these classes are considered as new units of a higher level of generalization than individuals and they allow to reduce the initial huge size of an input data set by summarizing it. Moreover, classes can represent real units that interest the data analyst. In this context, classes can be seen as a new data framework before extracting knowledge by data science methods and tools. The second step in SDA is to extend machine learning and statistical techniques to symbolic data.

The class framework provides some advantages (Diday 2016):

– When the population of study has million of individuals, the handling and study are facilitated summarizing it. From classes it is possible to summarize the data reducing the loss of information;
– Talking about confidentiality, classes can come up with a solution to preserve the individual's information stored once classes summarize the individuals;
– Transforming unstructured complex data in symbolic structured from this symbolic tools can be applied;

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– Due quantity of classes is mostly lower than individuals in an analysis, the interpretation of some data science methods becomes easier such as decision trees.

From the studies presented by Billard and Diday (2003, 2007) SDA has been widely used in statistics, data mining and machine learning accordingly showed Diday (2016, 2018). This progress is due to some characteristics:

– *Granularity changing* Considering that a researcher has information about the student (the grain), but he needs to understand the behavior of college (the class). The researcher can obtain college information through an aggregation process applied to student information. Then, different ways and applications can be explored;
– *Aggregation from different datasets* Based on the concept of classes two or more datasets can be used to compose only one dataset. This task can be made using the classes as a link key among the datasets. Then, important information coming from different places can be incorporated into the study;
– Data quality: SDA framework uses individuals in classes described by symbolic data, i.e., histograms, intervals, polygons, etc. This type of data makes possible to overcome common problems such as missing and noise data;
– *Internal variability and storage of information* Another important characteristic of SDA is that each of symbolic data presents variability in itself. The internal variability represents the dispersion of individuals in the class. In addition, symbolic data has better information storage capacity than classical data that use a centrality measure (mean, median etc) for describing individuals in the class.

Silva et al. (2019) introduced a new type of symbolic data, called symbolic polygonal data. To understand the behavior of this new type of symbolic data, the authors presented a variable named symbolic polygonal variable. For this variable, new data analysis measures were proposed, e.g., mean, covariance, variance, histogram, and so on. This new structure of symbolic data has essential properties:

– The construction of data is made based on the centrality and dispersion measures. Consequently, the descriptive measures for symbolic polygonal data are more accurate compared to those for interval data;
– Independent of number of sides symbolic polygonal data can be represented only two measures, center and radius. Then, an extensive dataset can be stored preserving their principal characteristics;
– Considering the aggregation task, the authors showed that when the number of sides of the polygon increases more variability is preserved that compared to interval data. This characteristic becomes the extraction of knowledge from symbolic data that is more robust.

The main goal of this paper is to present a general overview of the **psda** package (version 1.2.0) that implements descriptive and regression model based on least square estimation for symbolic poligonal data according to Silva et al. (2019) and to discuss the applicability of its functions. **psda** package provides ways to extract new knowledge from large data and represents this new knowledge through more complex data preserving internal variation. In addition, it opens up a vast domain for the development of new approaches in different domains such as supervised and non-supervised classification, time series for symbolic polygonal data in **R** language.

The first step in an analysis of symbolic data is building higher level units (here called classes) and obtaining their symbolic variables from a data set described by standard data. In SDA, it is very usual to describe classes by interval data since it is a kind of symbolic variable that expresses by itself a variability inside the class. In this case, each class is represented by the minimum and maximum values associated to each variable. However, this model of description based on position values only cannot model well the variability inside classes and produce loss of information.

Lima Neto and De Carvalho (2008) showed that the center and range measures of intervals are best way to model the information by symbolic interval variable in regression problems. Henceforth, the advances of SDA for regression models have considered center-range for representing interval data. In this context, the range of the intervals is adopted as dispersion measure for describing the classes and it overestimates sample dispersion. Thus, descriptive measures and modeling analyses for interval data cannot describe the real behavior of the data. It is known that in regression problems, the variability is a fundamental element for construction of the model estimates.

The use of symbolic polygonal variable (Silva et al. 2019) for describing classes aims to mitigate this problem of overestimation since the variability is measured by the sample dispersion of the data into each class. Section 2 presents some relevant works and packages of SDA in different domains. It can be seen that almost all works of SDA consider interval data. Section 3 presents the concept of the symbolic polygonal variable, knowledge extraction process, descriptive measures and linear regression model for symbolic polygonal data.

In order to evaluate the usefulness of the tool proposed in this work, we consider an application with Brazilian educational data. Here, the advantages of using polygonal data are:

– *Summarization of data* the Brazilian educational data is composed by two main datasets, *Student 2017* and *Cen-*

sus School containing 1.8 and 0.28 million observations described in $\mathbb{R}^4$ and $\mathbb{R}^2$, respectively. The datasets have a common variable named county. From symbolic polygonal approach we summarize both datasets without losing much information by county variable;
- *Combination of datasets* From aggregation task, it is possible to combine *Student 2017* and *Census School* data using the county variable as a connection key;
- *Interpretability Student 2017* and *Census School* contains information about student proficiencies and some infrastructure variables at Brazilian schools, respectively. From aggregation through county variable, we can analyze the behavior of proficiency students in a county (class). In this form, public policies can be realized in the counties to improve the performance of the students.

Section 4 shows an overview of the **psda** package, and Sect. 5 presents the application of the **psda** package with Brazilian educational data. Summary and discussion are presented in Sect. 6.

2 Theoretical advances and package of SDA

Theoretical developments and software have been done and continue in order to extend statistics and machine learning methods to symbolic data; but certainly, much remains to be done (Diday 2016). In the following, some relevant works and packages of SDA in different domains are described as follows. A more complete list of software for symbolic data can be found in Diday (2016), but some of them are deprecated.

2.1 Theoretical advances

Descriptive statistics Billard and Diday (2003) presented a detailed study about descriptive statistics to the univariate case using mean, variance, and histogram. To bivariate analysis, the authors present empirical joint density function and from it bivariate histogram, covariance, correlation, etc. Kao et al. (2014) enriched the exploratory data analysis to interval data from matrix visualization. The authors argue that the hypercubes (scatterplot to symbolic intervalar data) are limited until three dimensions, histograms do not provide extract knowledge from the interaction between variables, and zoom star (radar plot) is hard to interpret. They then use intervalar distances to build a matrix visualization that can be used to analyze multivariate variables and interactions.
Supervised classification Souza et al. (2011) developed four methods of classification to interval data based on logistic regression. This work uses different ways to represent the interval data. Araújo et al. (2014, 2016) developed classifiers for breast cancer detection; the authors used thermal photography. From this, they are building morphological and temperature matrices for each breast; then, they obtained the minimum and maximum values for each temperature matrix.
Non-supervised classification Topological clustering algorithms for interval data were proposed by Cabanes et al. (2013); De Carvalho et al. (2006) introduced a dynamical cluster for interval data based on L_2-distance; years later Irpino and Verde (2008) proposed a dynamical cluster based on Wasserstein distance; considering histogram symbolic variable (Irpino et al. 2014) discuss a clustering method using Wasserstein distance. Pimentel and Souza (2014) presented a multivariate fuzzy c-means method using interval variables to find the researcher's scientific production profiles in Brazil. For this, the authors developed a weight mechanism given by each variable incorporating its importance; this grows the robustness of the method. Irpino et al. (2017) proposed a fuzzy clustering of distributional data with automatic weighting of variable components.
Times series Maia et al. (2008) presented approaches to interval-valued time series forecasting. They use autoregressive, autoregressive integrated moving average (ARIMA), artificial neural network (ANN) and a combination of ARIMA and ANN models; a review of different techniques in forecasting for interval data is studied by Arroyo et al. (2010) and applied in financial data; Teles and Brito (2013) proposed to model interval time series with space-time autoregressive models.
Regression Billard and Diday (2002) proposed a first linear regression model using the lower and upper limits of the interval obtained from the classes. Posteriorly, Lima Neto and De Carvalho (2008) proposed a linear regression model through a new representation of interval data, named center and range. This approach shows significant advances in modeling of the interval data because it uses a centrality and dispersion measure to realize the fitting and prediction. Fagundes et al. (2014) proposed an interval kernel regression model; Outlier of a problem in regression is studied by Fagundes et al. (2013) with the robust regression model for interval data. Lima Neto and De Carvalho (2018) proposed an exponential type kernel robust regression model for interval variables based on the weighting of outliers observation. From this approach the authors present a method to understand the behavior of the outliers in the sample.
Applications Bezerra and De Carvalho (2010) proposed recommendation systems from three approaches based on the contend-based filtering, collaborative filtering, and hybrid filtering using symbolic multivalued variables. The methods consider a symbolic measure to compute the dissimilarity between multivalued variables represented by the weight distributions. Angadi and Kagawade (2017) used interval data to extract knowledge from different facial expression. The authors argue that the face recognition is a hard task with a minimum number of feature values. The two main steps real-

ized are: to cropped the facial image, and polar fast Fourier transformation (FFT) extraction (to build the intervalar data).

2.2 Packages

ISDA.R developed by Queiroz Filho and Fagundes (2012) is focused in interval data. The authors worked considering only descriptive measure for this type of data; in the package, it is possible to find an aggregation function, mean, variance, mode, histogram, 3D scatterplot, and so on. Actually, the package is in version 1.0 and can be downloaded from https://CRAN.R-project.org/package=ISDA.R.

RSDA introduced by Rojas et al. (2015) works with intervalar, modal, and histogram data. From this type of data, the authors implement descriptive measures, linear regression models as center and range (Lima Neto and De Carvalho 2008), and generalized linear models. Besides, diagnostic and error measures for these models are implemented as a determination coefficient; root-mean-squared error to lower and upper (RMSEL, RMSEU). Principal component analysis can be found in this package. The version is available in https://CRAN.R-project.org/package=RSDA, and the actual version is 3.0.4. It is essential to highlight the vignette of the package found in https://cran.r-project.org/web/packages/RSDA/vignettes/introduction.html.

GPCSIV, created by Brahim and Makosso-Kallyth (2013), implements an extension of the principal component analysis for interval data to handle big data. In the package, there are some interval datasets, for example: "Judge 1", "Judge 2", and "Judge 3" that contains information about agricultural products from six regions graded by three experts; "Òil" with eight observations and four interval variables (gravity, point, iodine value, and saponification). The current version is 0.1.0 and can be found in https://CRAN.R-project.org/package=GPCSIV

symbolicDA was developed by Dudek et al. (2015) and implemented a great number of the approaches in machine learning to symbolic data. Bagging, boosting, decision tree, Kohonen's self-organizing maps, principal component analysis, random forest, dynamic clustering, etc. The current version is 0.6-2 found in https://CRAN.R-project.org/package=symbolicDA.

3 Extracting knowledge from symbolic polygonal data

Figure 1 shows a workflow for building of the symbolic polygonal data and extracting knowledge from them. This process starts mining a classic data base through the choice of one or more categorical classic variables. These variables are used to summarize the classic data base and obtain new units called classes. Thus, the new data base is built where each row corresponds to a class (a sample of original units that have same categorical value) and the columns are bivariate representations of the classes by center and radius of the aggregated units. In the following, new variables, called symbolic poligonal variables, are defined for this new data base. Regression and descriptive statistics tools can be applied to symbolic poligonal variables.

3.1 Building symbolic polygonal data

Let $\mathcal{S}$ be a classical sample space with n items described by p classical variables $\mathcal{X} = \{X_1, \ldots, X_p\}$. Classes of individuals of $\mathcal{X}$ can be obtained by clustering process or aggregation method from at least a categorical variable. In this work, we consider the classes are obtained from categorical variables.

Let E be a symbolic sample with $m < n$ items described by symbolic variables $\mathcal{Z} = \{Z_1, \ldots, Z_q\}$, where $q < p$. Given a categorical variable X_t of $\boldsymbol{X}$, a class is defined for all individuals of $\mathcal{S}$ that has a particular category assumed by the variable X_t. In this context, each class is in high granularity and it is composed by a subset of individuals that are in lower granularity.

A polygonal variable Z is a correspondence from a set E into $\mathbb{R}^2$ which is graphically represented for all $i \in E$, the vertices of $Z(i) = \{(a_{i1}, b_{i1}), \ldots (a_{i\ell}, b_{i\ell})\}$ form a convex polygon in $\mathbb{R}^2$, where $a_i, b_i \in \mathbb{R}$, $\ell = 1, \ldots, L$, and $L \in \mathbb{N} \geq 3$. The bound is obtained from intersection of finitely many half-plane of the form $\{(x, y) \in \mathbb{R}^2 : \alpha x + \beta y < c\}$ for $\alpha, \beta, c \in \mathbb{R}$. Besides, the points of the intersection of each half-plane are called vertices.

Now, a polygon is represented by its center and radius with L sides. To illustrate the construction of symbolic polygonal data a simple example is considered. The score of six students with 7 and 8 years old is shown in Table 1.

To obtain the polygonal variable, we aggregate the student's scores by years old to build the classes 1 and 2 for 7 and 8 years old, respectively. For this, each polygon is represented for its center and radius using the mean and two times the standard deviation of students for each year old, respectively. Table 2 displays the polygons that describe the classes 1 and 2.

In this example, we assume $L = 5$ and one pentagon is constructed for each class based on Eq. (1),

$$P_{j\ell} = (a_{j\ell}, b_{j\ell}) = \left(c_j + r_j \cos\left(\frac{2\pi\ell}{L}\right), c_j + r_j \sin\left(\frac{2\pi\ell}{L}\right)\right), \quad (1)$$

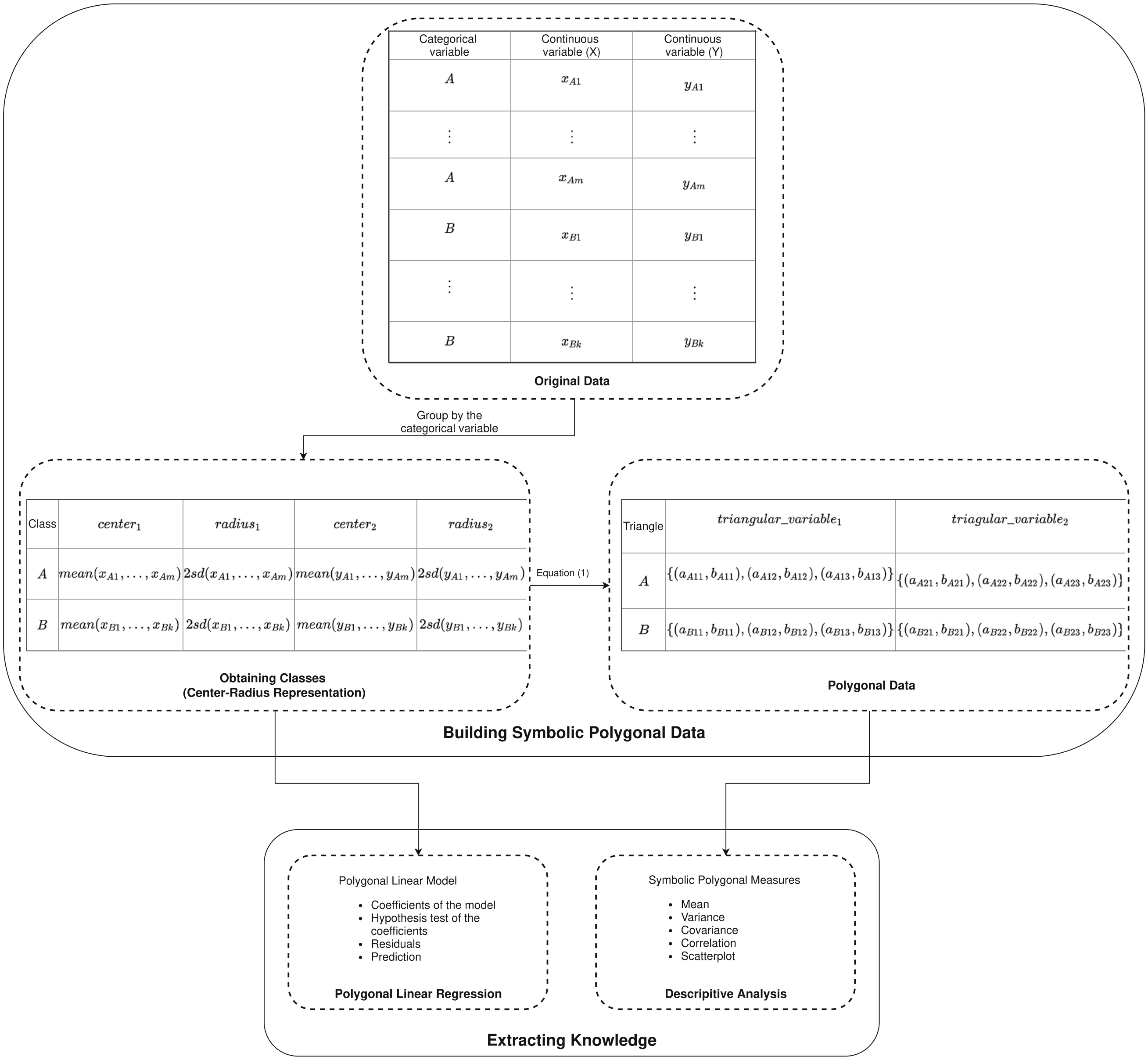


Fig. 1 Extracting and mining knowledge for symbolic polygonal data

Table 1 Example of classical data

Student	Years old	Score (X)
1	7	4.75
2	7	4.77
3	7	6.02
4	8	10.88
5	8	9.25
6	8	10.80

where j indicates the number of classes, c_j and r_j are the center and radius, respectively, of the polygon in class j, and $\ell = 1, \ldots, L$.

Table 2 Polygonal data represented by center and radius

Class	Score	
	Center	Radius
1	5.18	1.46
2	10.31	1.84

Figure 2 illustrates the score of students in classes 1 and 2. We can observe that the average score of class 1 is lower than that in class 2 by the polygon positions. Besides, class 2 has more variability than class 1; this can be seen from the size of polygon. In other words, a large polygon indicates more the variability into class.

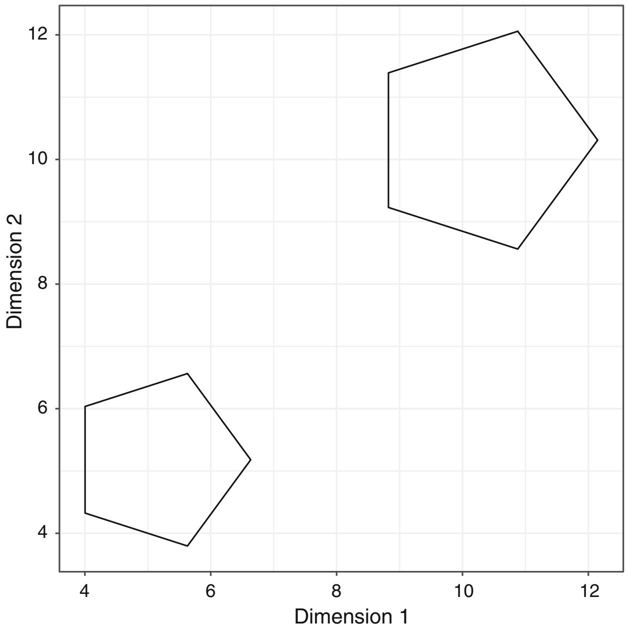


Fig. 2 Symbolic polygonal scatterplot for score variable

3.2 Extraction of knowledge

The extraction knowledge step can be performed from symbolic polygonal descriptive measures, e.g., mean, covariance, variance, correlation coefficient and so on. These measures are fundamental to understand the behavior of the variables involved in the problem and they can define the best strategy to build regression models, time series, etc.

Let $E = \{1, \dots, m\}$ be a set of m objects described by a symbolic polygonal variable $Z = (Z_1, Z_2)$, where the components Z_1 and Z_2 are random variables that describe the first and second dimensions, respectively. Different to a mean of classical variable, the mean of a polygonal variable $\overline{Z} = (\overline{Z}_1, \overline{Z}_2)$ is a bidimensional value that represents a centrality measure of a polygonal distribution composed of m objects. For $m = 1$, the mean of Z is the center coordinates of the polygon.

The measure $\overline{Z}$ is given by

$$\overline{Z} = \left(\frac{1}{6m} \sum_{i \in E} \sum_{\ell=1}^{L} \frac{(a_{i,\ell} + a_{i,\ell+1}) c_{i,\ell}}{A_i}, \right. \\ \left. \frac{1}{6m} \sum_{i \in E} \sum_{\ell=1}^{L} \frac{(b_{i,\ell} + b_{i,\ell+1}) c_{i,\ell}}{A_i} \right), \tag{2}$$

where $c_\ell = a_{i,\ell} b_{i,\ell+1} - a_{i,\ell+1} b_{i,\ell}$, and

$$A_i = \frac{1}{2} \sum_{i \in E} \left| \sum_{\ell=1}^{L} b_{i,\ell} (a_{i,\ell+1} - a_{i,\ell-1}) \right|. \tag{3}$$

for $a_{i,L+1} = a_{i,1}, a_{i,0} = a_{i,L}, b_{i,L+1} = b_{i,1}$, and $b_{i,0} = b_{i,L}$. $(a_{i,\ell}, b_{i,\ell})$ are the vertices of ith polygon.

The variance of a polygonal variable $S^2(Z) = (S_1^2, S_2^2)$ is based on the second empirical moment of Z given by $M2(Z) = (M2_1, M2_2)$ where $M2_1$ is the second empirical moment of the first coordinate and $M2_2$ is the second empirical moment of the second coordinate.

$$S(Z) = \left(S_1^2, S_2^2\right) = \left(M2_1 - \overline{Z}_1^2, M2_2 - \overline{Z}_2^2\right) \tag{4}$$

where

$$M2_1 = \frac{1}{12m} \sum_{i \in E} \sum_{\ell=1}^{L} \frac{(a_{i,\ell}^2 + a_{i,\ell} a_{i,\ell+1} + a_{i,\ell+1}^2) c_{i,\ell}}{A_i}, \tag{5}$$

$$M2_2 \frac{1}{12m} \sum_{i \in E} \sum_{\ell=1}^{L} \frac{(b_{i,\ell}^2 + b_{i,\ell} b_{i,\ell+1} + b_{i,\ell+1}^2) c_{i,\ell}}{A_i}. \tag{6}$$

Moreover, the covariance $\text{Cov}(Z_1, Z_2)$, correlation $r(Z_1 Z_2)$ and coefficient of variation $\text{CV}(Z)$ are defined, respectively, by

$$\text{Cov}(Z_1, Z_2) = M_{12} - \overline{Z}_1 \, \overline{Z}_2,$$
$$r(Z_1 Z_2) = \frac{\text{Cov}(Z_1, Z_2)}{\sqrt{S_1^2 \times S_2^2}} \text{ and}$$
$$\text{CV}(Z) = \frac{\sqrt{S_1^2 S_2^2}}{\overline{Z}}.$$

where $M2_{12}$ is the second joint moment of Z

$$M2_{12} = \frac{1}{24m} \sum_{i \in E} \sum_{\ell=1}^{L} \\ \frac{(a_{i,\ell} b_{i,\ell+1} + 2a_{i,\ell} b_{i,\ell} + 2a_{i,\ell+1} b_{i,\ell+1}) c_\ell}{A_i} \tag{7}$$

In addition to descriptive measures, a data scientist needs to build statistical models that can understand the influence of explanatory variables under a dependent variable. This can be performed through regression models, e.g., the linear regression. Regarding symbolic polygonal data, a model is described as follows.

Given a set of m aggregated units (classes) obtained by an aggregation process, concatenation and fusion. Each unit is described by a dependent polygonal variable Y and p regressors polygonal variables $\{X_1, \dots, X_p\}$. Each polygonal variable is represented by a pair of coordinates: center and radius.

The polygonal linear regression model is defined as $\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon}$, where the vector of dependent variable is $\boldsymbol{y} = (\boldsymbol{y}_c^T, \boldsymbol{y}_r^T)^T$ for $\boldsymbol{y}_c = (y_{c1}, \dots, y_{cm})^T$ and $\boldsymbol{y}_r =$

$(y_{r1}, \ldots, y_{rm})^T$, the unknown parameters vector is given by $\boldsymbol{\beta} = (\boldsymbol{\beta}_c^T, \boldsymbol{\beta}_r^T)^T$ for $\boldsymbol{\beta}_c = (\beta_{c1}, \ldots, \beta_{cp})^T$ and $\boldsymbol{\beta}_r = (\beta_{r1}, \ldots, \beta_{rp})^T$, the error vector is $\boldsymbol{\varepsilon} = (\boldsymbol{\varepsilon}_c^T, \boldsymbol{\varepsilon}_r^T)^T$ for $\boldsymbol{\varepsilon}_c = (\varepsilon_{c1}, \ldots, \varepsilon_{cm})^T$ and $\boldsymbol{\varepsilon}_r = (\varepsilon_{r1}, \ldots, \varepsilon_{rm})^T$. The matrix of the model is given by

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{X}_c & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{X}_r \end{bmatrix}, \tag{8}$$

where $\boldsymbol{0} = (0_{j1}, \ldots 0_{jp})$, $\boldsymbol{X}_c = (x_{cj1}, \ldots, x_{cjp})$, $\boldsymbol{X}_r = (x_{rj1}, \ldots, x_{rjp})$ and $x_{cj1} = x_{rj1} = 1$ for $j = 1, \ldots m$.

Using least squares method, the solution for the vector of unknown parameters is given by $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_{\boldsymbol{c}}{}^T, \hat{\boldsymbol{\beta}}_{\boldsymbol{r}}{}^T)^T = (\boldsymbol{X}^{\boldsymbol{T}} \boldsymbol{X})^{-1} \boldsymbol{X}^{\boldsymbol{T}} \boldsymbol{y}$.. The predicted polygon $\hat{P}_j$ $(j = 1, \ldots, m)$ with L vertices is obtained by

$$\hat{P}_{jl} = \left(\hat{y}_{cj} + \hat{y}_{rj} \cos\left(\frac{2\pi\ell}{L}\right), \hat{y}_{cj} + \hat{y}_{rj} \sin\left(\frac{2\pi\ell}{L}\right) \right), \tag{9}$$

for $\ell = 1, \ldots, L$.

If $\hat{y}_{rj} < 0$ was considered $\hat{y}_{rj} = 0$ $(j = 1, \ldots, m)$. In this case, the predicted polygon is degenerated, i.e., a point. In addition, the polygonal linear regression presents an unique residual, given by $\hat{\boldsymbol{\varepsilon}} = \boldsymbol{y} - \hat{\boldsymbol{y}}$.

Algorithm 1: Linear regression model for polygonal symbolic data

Data: symbolic polygonal dataset with L vertices
Result: residuals ($\hat{\boldsymbol{\varepsilon}}$), predicted polygon ($\hat{P}$) with L slides

```
begin
    m = sample size ;
    for j = 1 : m do
        compute y_cj using Eq. (2) and y_rj is given by
        euclidean distance between center of response
        variable and any vertex of polygon;
    end
    for j = 1 : m v = 1 : p do
        compute x_cjv using Eq. (2) and x_rjv as the Euclidean
        distance between center of predictor variable v and
        any vertex of polygon v;
    end
    compute β̂ = (β̂_c^T, β̂_r^T)^T = (X^T X)^{-1} X^T y ;
    compute ŷ = β̂X ;
    for j = 1 : m do
        if ŷ_rj < 0 then
            ŷ_rj = 0;
        end
    end
    compute ε̂ = y − ŷ;
end
```

Table 3 List of **psda** functions and their main characteristics

Name	Outcome	Type of function
`parea`	Double	Auxiliary
`pconvex`	Boolean	Auxiliary
`paggreg`	Data frame	Auxiliary
`spolygon`	Matrix	Auxiliary
`psymbolic`	Environment	Auxiliary
`na.omit`	Environment	Auxiliary
`pcorr`	Vector	Descriptive
`pcov`	Vector	Descriptive
`pmean`	Vector	Descriptive
`pmean_id`	Vector	Descriptive
`pplot`	Graphic	Descriptive
`psim`	Polygonal	Descriptive
`psmi`	Vector	Descriptive
`pvar`	Vector	Descriptive
`pvari`	Vector	Descriptive
`plr`	Environment	Modeling
`summary.plr`	Environment	Modeling
`print.plr`	–	Modeling
`print.summary.plr`	–	Modeling
`rmsea`	Double	Modeling

4 Overview of the psda package

The **psda** provides functions to symbolic polygonal data analysis using descriptive measures, and a polygonal linear regression model. To install the latest release version of **psda** from CRAN we can use the installation in R `install.packages("psda")`. The current development version can be downloaded manually in https://cran.r-project.org/package=psda or from https://github.com/wagnerjorge/psda.

There are 20 functions classified in descriptive, modeling and auxiliary measure. The Auxiliary functions describe the mathematical properties, the building of symbolic polygonal data, and are used to facilitate the implementation of descriptive and model functions. Besides, the descriptive and modeling functions are used to extract knowledge from polygonal data and are the mathematical equations presented in Sect. 3. Details can be seen in Table 3.

4.1 Auxiliary functions

1. `pconvex` requires `polygon` as argument a matrix of dimension $\ell \times 2$, where ℓ and 2 represent the number of vertices and polygon dimension, respectively. It verifies if the polygon is convex and results in a boolean, `TRUE` if the polygon is convex and `FALSE` otherwise;

2. `paggreg` aggregates the classical data obtained from p variable. The aggregation is made through center and radius representation as shown in Eq. (1). The result is `paggregated` an object containing a `data.frame` for the center and other for radius considering $p > 1$. For $p = 1$ two vectors are the output one for center and other for radius.
3. `spolygon` allows to obtain a single symbolic polygonal. It uses as parameters the `center`, `radius` and `vertices`. The output is a matrix with l rows and 2 columns that represent the number of vertices and the dimension, respectively. `psymbolic` uses an object of the class `paggregated` and `vertices` to obtain symbolic polygonal variables distributed in p variables, where p is the number of classical variables.
4. `parea` is the implementation of Shoelace equation described in Eq. (3). For calculating the area from `parea` it is needed to guarantee that the polygon is convex, then can be used the function `pconvex`. The argument of the function is `polygon`, a matrix of dimension $\ell \times 2$, where ℓ and 2 represent the number of vertices and the dimension of the polygon, respectively.
5. In symbolic polygonal missing presence, the data scientist should use the function `na.omit`. It removes the symbolic individual that is missing, NA.

4.2 Descriptive measure functions

1. Two functions are used to calculate the symbolic polygonal mean as described by Eq. 2; they are `pmean_id` and `pmean`. The first function, `pmean_id`, calculates the symbolic polygonal mean when $m = 1$, i.e., there is a single polygon. This means it is known as symbolic internal polygonal mean. The argument required is `polygons` equal to `pconvex` and `parea`. The second function, `pmean`, calculates the symbolic polygonal mean considering a symbolic polygonal variable, i.e., the observations are polygons. For this function the input is `polygons`. In addition, the output for `pmean_id` and `pmean` is a bi-dimensional vector with the barycenter of polygon and the mean of barycenter, respectively;
2. The symbolic polygonal second moment for one polygon, i.e., $m = 1$ is calculated by `psmi`. A single polygon is needed and the outcome is a bi-dimensional vector. The function is used to calculated the symbolic polygonal internal variance ($m = 1$), `pvari`. For both function the parameter is `polygon` a matrix of dimension $\ell \times 2$. The function `pvar` is used to calculate the symbolic polygonal variance. The input parameter is a list of polygons `polygons`, and the outcome is a bi-dimensional vector containing the dispersion of a symbolic polygonal variable;
3. Symbolic polygonal covariance and correlation are calculated from implementation of `pcov` and `pcorr`, respectively. The input parameter of the both functions is `polygons`, a list of polygons and each element of the list is a matrix of dimension $\ell \times 2$. The outcome is an integer;
4. Regarding the interest in simulating a symbolic polygonal variable composed of $m \in \mathbb{N}$ individuals and ℓ vertices, the research can use the function `psim`. The function is the implementation of Eq. (1) and requires the number of polygons and the number of vertices, `objects` and `vertices`, respectively. The output is a list of simulated polygons;

4.3 Modeling function

`plr` is the implementation of the polygonal linear regression model presented in Sect. 3. The input parameters required are:

1. `formula` an object of class `formula`, for example, if was considered one dependent an three regressors variables, the formula can be described as `y ~x1 + x2 + x3`.
2. Each symbolic polygonal variable is described by a list containing the polygons that represent the observation. Then, `data` is an environment containing all symbolic polygonal variables of the model;
3. `...` are ellipses that describe additional arguments that can be passed to polygonal linear regression fitting functions.

The `plr` outputs are:

1. `residuals` is a vector of residuals calculated as described in polygonal linear regression at Sect. 3;
2. `rank` is the dimension of matrix model. If all variables are independent rank $= 2p$, where p is the number of regressors, if there is dependence between two variables rank $= 2(p - 1)$;
3. `call` return an object of the class `call` with the matched call;
4. `fitted.values` a matrix $2n \times 1$, where n indicates the number of samples. The first n positions are the $\hat{y}_c$, the others are $\hat{y}_r$, center and radius estimates, respectively.
5. `terms` represents an environment with objects of the class `terms` and `formula`;
6. `coefficients` accessed the estimative of the parameters . Regarding the `intercept = TRUE` the variable is given by vector of length $2p + 2$, otherwise the vector presents length $2p$;

7. The matrix model given by Eq. (8) is given by `model` output.

The main information about the model can be obtained through method `summary` of the class `plr`. These measures are fundamentals to understand the characteristics of the model, for example, residuals and its variance, the coefficients and so on. The inputs of the method are:

1. `x` an object of the class `plr`, in general, it is a call of `plr`;
2. The other input are ellipses `...` that represent further arguments passed to or from other methods.

The outputs of the method are:

1. `residuals` a vector $2m$ calculated as $\boldsymbol{y} - \hat{\boldsymbol{y}}$;
2. `aliased`, a named logical vector showing if the original coefficients are aliased;
3. Similarly the `term` in `plr` function, the `summary` method returns its `term`, an environment with objects of the class `terms` and `formula`;
4. A $p \times 4$ matrix with columns for the estimated coefficients and its standard error, z statistic and corresponding (two-sided) p value are given by output `coefficients`.

To print the `plr` model in default of R the method `print` of the class `plr` was developed. Similarly, the method `print` of the `summary.plr` is developed. Both methods do not present output; yours input parameters are the same and given by:

1. `x` an object of the class `plr`, it is a call of `plr`;
2. `digits` is a non-null value for digits; it specifies the minimum number of significant digits to be printed in values;
3. `...`, the ellipses are used as further arguments passed to or from other methods.

The root-mean-squared error of area (ϕ) proposed by Silva et al. (2019) is a performance measure, and it was implemented in **psda** as function `rmsea`. The input is the `observed` and `fitted`, the observed polygons, and fitted polygons, respectively. The output is the measure ϕ.

5 An application with Brazilian educational data

psda package is applied to Brazilian educational data in order to illustrate the processes of building symbolic polygonal data and extracting knowledge from these new data. Moreover, a model to estimate mathematical proficiency of Brazilian students in the final year of elementary education is proposed. For this, two datasets are connected to obtain a single dataset. First contains information about the SAEB and second contains infrastructure of the schools.

Brazil occupies the last positions in the education theme according to the Programme for International Student Assessment (Pisa) rank. Then, approaches for understanding the main factors that exercise influence on Brazilian public politics to improve the education of the country should be encouraged. For this reason, we applied some tools of the psda package in order to extract new knowledge with symbolic polygonal data and give a new vision of the problem.

The *Student 2017* dataset contains 1.8 millions of observations and 3 variables `county`, `proficiency_lp`, and `proficiency_mt` that mean county, Portuguese language proficiency, and mathematical proficiency, respectively. These data are built from student information on the final year of elementary education in 2017. The *Census School* dataset contains 0.28 millions of observations and 5 variables `county`, `classroom`, `classroom_used`, `computers`, `employees` that it indicates the number of classroom, number of classroom used, computer numbers and functionary numbers, respectively. The datasets are obtained from Brazilian Institute of Studies and Educational Research Anísio Texeira (INEP) through the site http://inep.gov.br/dados.

An important characteristic of using symbolic data is the capacity of combining many datasets from a common variable. Here, both original data sets are described by the `county` variable and this variable allowed to connect the two datasets to form a single dataset. In the following, the aggregation process is carried out regarding the `county` variable. Thus, classes of individual units are obtained and the original dataset is summarized. The new dataset has 4037 items described by 7 polygonal variables. Given a variable, each polygonal data is represented by the center and radius of the polygonal. The classes datasets for center and radius are formed, respectively, and they can be found in https://github.com/wagnerjorge/brstprof_psda.

In order to support the knowledge extraction step through regression model, the dataset of classes is partitioned randomly into two subsets: train (70% of items) for building the model and test (30% of items) for validating the model. Moreover, a seed is used to preserve the reproducibility of this partitioning process.

5.1 Descriptive measures for Brazilian educational data

"Appendix A" displays some descriptive measures for poligonal data such as mean, variance, covariance, and correlation using the `pmean`, `pvar` and `pcorr`, respectively. The mean, standard deviation and correlation are, respectively: (249.168 249.168), (47.764, 47.764) and 0.006 for mathematical pro-

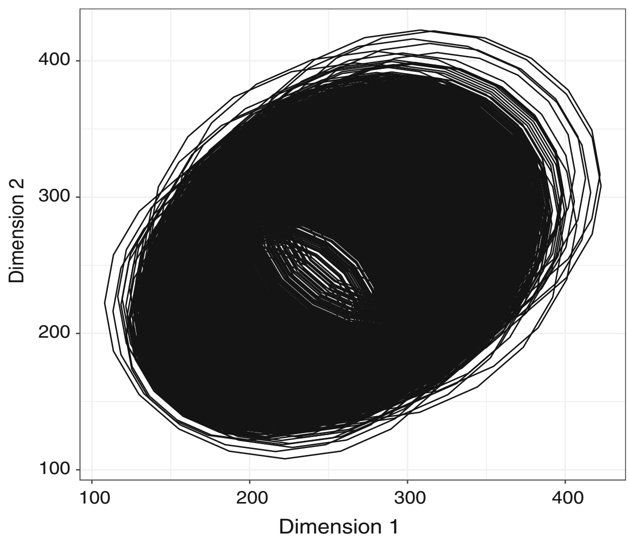


Fig. 3 Symbolic scatterplot of mathematical proficiency variable

ficiency variable; (249.219 249.219), (47.739, 47.739) and 0.006 for Portuguese proficiency variable; (27.366 27.366), (30.402, 30.402) and 0.000 for employees variable.

psda provides a function to build a symbolic polygonal scatterplot, named `pplot`. Besides, "Appendix A" shows details about the code to create the symbolic scatterplot for the variable `proficiency_mt`. Figure 3 shows the symbolic scatterplot. In this figure, it is possible to see that the polygons present an increasing linear relation and the icosagons present different radius.

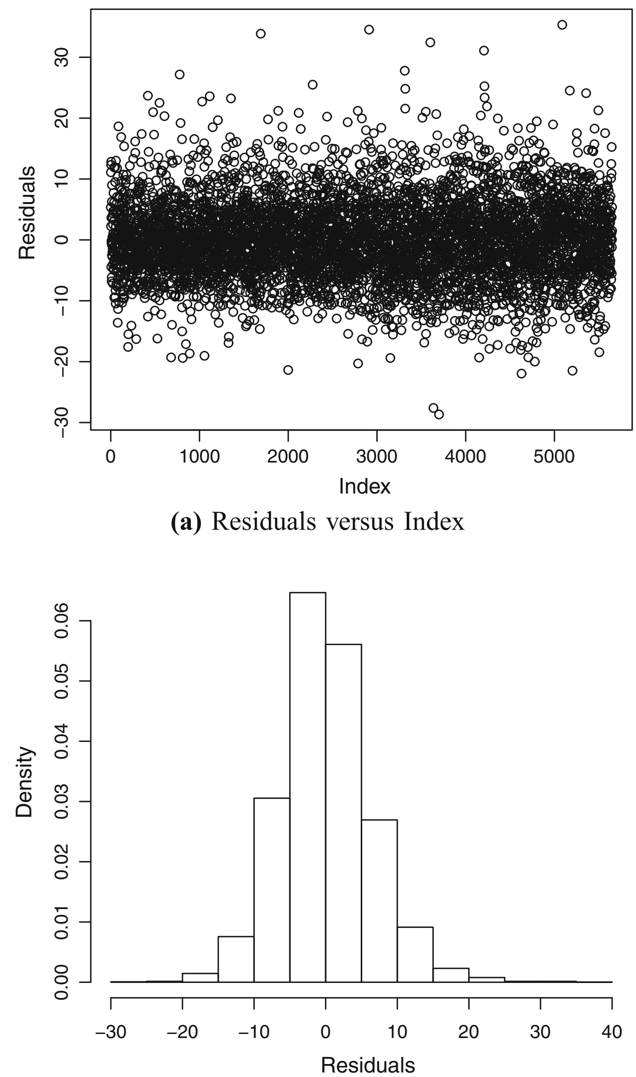
5.2 Polygonal linear regression with psda

The polygonal linear regression model is built from `plr` function. The variables of the regression problem are: in `proficiency_mt` as response variable and the regressor variables `proficiency_lp, classroom_used, computers, classroom`, and `employees`. The function `plr` produces the parameter estimates of the model: $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_c^T, \hat{\boldsymbol{\beta}}_r^T)^T$.

As example of parameter interpretation for center data model, we have that β_{c1} represents the difference in the predicted value of mathematical proficiency for each one-unit difference in Portuguese language proficiency, if the others regressors remain constants.

A scatterplot and a histogram of the residuals can be accessed by `fit$residuals` as it is shown in "Appendix A". The models present other elements that can be displayed from `names(fit)`.

Figure 4 shows the behavior of the residuals for the polygonal regression model. From Fig. 4a we can observe that the residuals are randomness around the value zero. The



(a) Residuals versus Index

(b) Histogram

Fig. 4 Scatterplot of the residuals versus index and histogram of residuals obtained from polygonal linear regression (PLR) model

histogram of residuals in Fig. 4b reveals an approximately normal distribution with mean zero.

The main information about the PLR model can be seen through `summary` of the model. This function is a method of the class `plr` and needs of the object `fit`. Moreover, it returns the values: `coefficients`, `residuals`, `sigma`, `aliased`, `call`, `terms`.

"Appendix A" shows the code to obtain the followings results of the summary that are: estimates of the unknown parameters, the hypothesis test of parameters and your standard error, z statistic, p value, and significance of the model. In this application, regarding the center data, the intercept, Portuguese language proficiency, and computer numbers are significant predictors for the model. Similarly, the radius

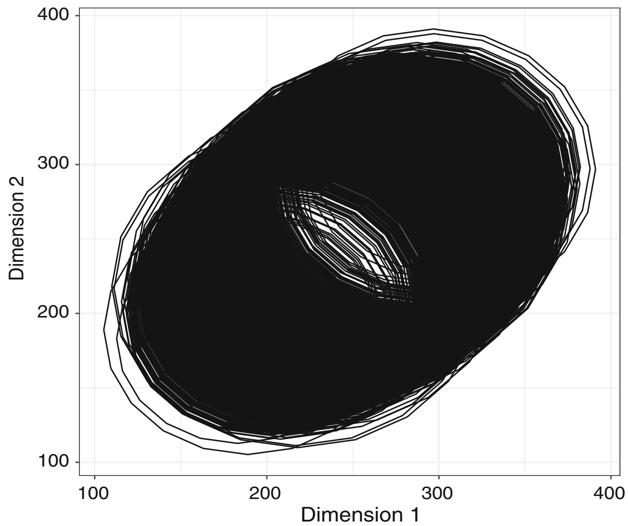


Fig. 5 Symbolic polygonal scatterplot of mathematical proficiency

data, the intercept, Portuguese language proficiency, and computer numbers are significant predictors.

From `predict` command, it is possible to obtain the fitted polygons in center and radius data. When we use `polygon = T` as argument of `predict`, it is possible to build the fitted polygon representing the data.

From `pplot` function, we can display the symbolic polygonal scatterplot for fitted mathematical proficiency data (`fitted_polygons`). Figure 5 shows the symbolic polygonal scatterplot for fitted variable `proficiency_mt`. We can observe that the fitted icosagons have linear behavior and different radius sizes.

Finally, the root squared mean error for the polygonal area can be obtained through function `rmsea` that is used to verify the performance measure of the polygonal linear model.

5.3 Prediction of mathematical proficiency

Given test dataset and coefficients of the polygonal linear regression accessed from `fit$coefficients`, the predicted polygons are obtained using the predictor matrix according to Eq. (8).

"Appendix A" displays too commands to build the predicted polygons based on the center and radius data for mathematical proficiency and the `psymbolic` function with `v = 20`. Besides, for this application the performance measure for the fitted model is RMSEA $= 4106.117$.

To illustrate the prediction given by the PLR model, two polygonal data are obtained from the test dataset. The polygons have $\mathrm{id}_{\mathrm{test}}$ equal to 1,100,015 and 1,100,056 and represent the counties Cerejeiras and Alta Floresta D'Oeste. The polygonal means for Cerejeiras and Alta Floresta D'Oeste are: (259.88, 259.88) and (273.97, 273.97, respectively. It indicates that the students of the county Alta Floresta D'Oeste present proficiency higher than students of the Cerejeiras. The polygonal standard deviations are (16.12, 16.12) for Cerejeiras, and (16.55, 16.55) for Alta Floresta D'Oeste (see, "Appendix A").

In order to compare the performance of the PLR model presented in this paper with other models of the SDA literature, models for interval data are adopted since that squares can be seen as a particular polygon. In this context, the center and range model (CRM) introduced by Lima Neto and De Carvalho (2008) and exponential-type kernel robust regression for interval-valued variables (iETKRR) proposed by Lima Neto and De Carvalho (2018) are considered in this evaluation. These models were chosen because they are linear models as PLR one.

The experimental evaluation is carried out regarding hold-out method in which the data set is divided by train (70%) and test (30%) data sets randomly. In addition, 100 replications of this procedure are obtained in the experience Monte Carlo framework. The performance of the PLR, CRM, and iETKRR models is measured by the root-mean-squared error of area (RMSEA), and Wilcoxon signed-rank tests are applied to compare the PLR versus CRM and iETKRR.

Given a test dataset of size r, the performance of three models can be measured by the root-mean-square error of area given by

$$\phi = \sqrt{\frac{1}{r}\sum_{u=1}^{r}\left[\mathrm{area}(\hat{P}_u) - \mathrm{area}(P_u)\right]^2}. \tag{10}$$

where $\hat{P}$ is the predicted polygon and P is the observed polygon.

Figure 6 shows boxplots for RMSEA metric with respect to 100 test datasets and PLR, CRM and iETKRR models. From this figure, we can see clearly that PLR presents the best performance compared to CRM and iETKRR models. Moreover, Wilcoxon signed-rank test at significance level $\alpha = 0.05$ confirms the superiority of the PLR model in comparison with CRM and iETKRR ones with p values < 0.0001.

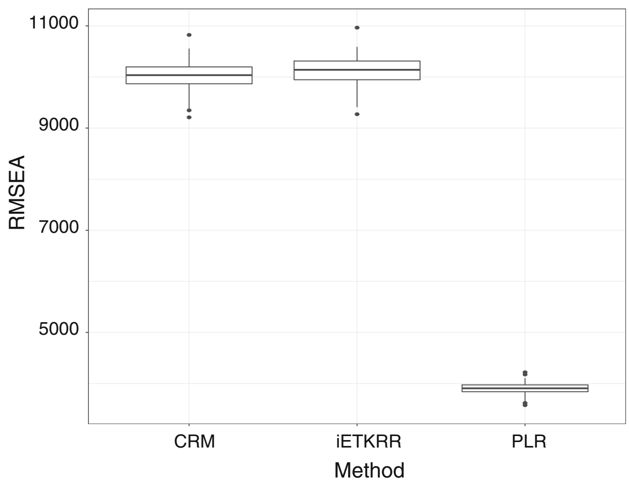


Fig. 6 Performance of the methods in test data selected from a hundred Monte Carlo subsamples

6 Summary and discussion

This paper showed a package for handling symbolic data based on a new type of data, called symbolic polygonal data. It starts with a step of knowledge extraction from classical datasets in order to obtain symbolic polygonal data. This new type of data is represented by the center and radius of the polygon and it is an efficient way to: data storage without significant loss of information, less sensitive performance in the outlier and missing data presence. The theoretical background for polygonal data is discussed and an application of real world is presented in order to show the usefulness of the proposed package.

The `psda` R package involves functions for extracting knowledge from symbolic polygonal data regarding descriptive measures and a regression model. It highlights an important advances in the vanguard of data analysis in various research domains, e.g., statistics, data mining and machine learning. The `psda` package focuses descriptive measures, scatterplot and a regression model for symbolic polygonal data. It opens ways for new descriptive measures and the construction of new models in the statistical and machine learning areas for polygonal data.

6.1 Computational details

The results in this paper were obtained using R 3.3.3 with the auxiliary packages **ggplot2**, **rgeos**, **plyr**, **sp**, **plot3D** and **raster**. R itself and all packages used are available from the Comprehensive R Archive Network (CRAN) at https://CRAN.R-project.org/.

ggplot2 is an important package to plot the graphics, in especial, the symbolic polygonal scatterplot `pplot`. The auxiliary packages **rgeos**, **raster** and **sp** were used to calculate the intersection of polygons. The manipulation of `data.frame` was done from functions of package **plyr**.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Appendix

This section details all results obtained in SAEB 2017 application from **psda**.

```
library(psda)
library(ggplot2)

data(saeb2017)
head(saeb2017)

center <- data.frame(county = saeb2017$county, saeb2017[, 2 : 7])
radius <- data.frame(county = saeb2017$county, saeb2017[, 8 : 13])

names(center) <- gsub("\\_center.*","", names(center))
names(radius) <- gsub("\\_radius.*","", names(radius))

head(round(center, 1))

    county proficiency_lp proficiency_mt classroom classroom_used computers employees
1: 1100015          258.1          258.1       6.8            7.5       6.4      26.6
2: 1100023          262.8          261.1      13.3           11.8      28.7      59.6
3: 1100049          259.5          260.8      12.0           10.4      22.5      38.0
4: 1100056          271.3          276.8       9.8            8.7      13.9      33.2
5: 1100064          255.5          257.4      13.6           11.7      29.8      49.1
6: 1100072          251.1          249.7       7.4            7.4      11.7      33.0

head(round(radius, 1))
    county proficiency_lp proficiency_mt classroom classroom_used computers employees
1: 1100015           89.0           83.7      11.7           16.8      26.2      56.3
2: 1100023           85.1           82.1      12.3           12.9      72.1      63.6
3: 1100049           91.6           88.1      15.0           12.9      57.7      57.2
4: 1100056           84.1           79.4       8.0            8.3      28.0      43.3
5: 1100064           91.3           90.3      16.8           16.9     110.0     115.6
6: 1100072           85.7           81.4       4.9            5.6      25.8      25.2

RNGversion('3.5.3')
set.seed(1)
samp <- sample(1:nrow(center),
.7*nrow(center), replace = F)

center_train <- center[samp, -1]
radius_train <- radius[samp, -1]

center_test <- center[-samp, -1]
radius_test <- radius[-samp, -1]
n_test <- nrow(center_test)
id <- center$county
id_test <- id[-samp]

dta_train <- list()
dta_train[[1]] <- center_train
dta_train[[2]] <- radius_train
names(dta_train) <- c('center', 'radius')
class(dta_train) <- 'paggregated'

dta_test <- list()
dta_test[[1]] <- center_test
dta_test[[2]] <- radius_test
names(dta_test) <- c('center', 'radius')
class(dta_test) <- 'paggregated'

v <- 20
pol_dta_train <- psymbolic(dta_train, v)
pol_dta_test <- psymbolic(dta_test, v)
```

```
head(pol_dta_train$proficiency_mt, 1)
[[1]]
       [,1]      [,2]
[1,] 333.9484 269.2740
[2,] 319.6404 297.3551
[3,] 297.3551 319.6404
[4,] 269.2740 333.9484
[5,] 238.1458 338.8786
[6,] 207.0176 333.9484
[7,] 178.9365 319.6404
[8,] 156.6512 297.3551
[9,] 142.3432 269.2740
[10,] 137.4130 238.1458
[11,] 142.3432 207.0176
[12,] 156.6512 178.9365
[13,] 178.9365 156.6512
[14,] 207.0176 142.3432
[15,] 238.1458 137.4130
[16,] 269.2740 142.3432
[17,] 297.3551 156.6512
[18,] 319.6404 178.9365
[19,] 333.9484 207.0176
[20,] 338.8786 238.1458

pmean(pol_dta_train$proficiency_mt)
[1] 249.1677 249.1677

pmean(pol_dta_train$proficiency_lp)
[1] 249.2189 249.2189

pmean(pol_dta_train$employees)
[1] 27.36636 27.36636

sqrt(pvar(pol_dta_train$proficiency_mt))
47.76465 47.76465

sqrt(pvar(pol_dta_train$proficiency_lp))
47.73871 47.73871

sqrt(pvar(pol_dta_train$employees))
[1] 30.40192 30.40192

pcorr(pol_dta_train$proficiency_mt)
[1] 0.005554888

pcorr(pol_dta_train$proficiency_lp)
[1] 0.005556134

pcorr(pol_dta_train$employees)
[1] 0.000268102


pplot(pol_dta_train$proficiency_mt) + labs(x = 'Dimension 1', y = 'Dimension 2') +  theme_bw()
```

```

fit <- plr(proficiency_mt ~ proficiency_lp + computers +
            classroom + employees + classroom_used, data = pol_dta_train)
fit

Call:
plr(formula = proficiency_mt ~ proficiency_lp + computers + classroom +
     employees + classroom_used, data = pol_dta_train)

Coefficients:
(center-intercept) center-proficiency_lp center-computers center-classroom  center-employees
-8.977193               1.037758          0.087790         -0.153068         -0.024288
center-classroom_used (radius-intercept) radius-proficiency_lp radius-computers
0.065484           24.115389              0.702786          0.027207
radius-classroom radius-employees radius-classroom_used
0.009482         0.001955              0.006586

plot(fit$residuals, ylab = 'Residuals')
hist(fit$residuals, xlab = 'Residuals', prob = TRUE, main = '')

names(fit)
[1] "na.action" "residuals" "rank"  "call"
[5] "fitted.values" "terms" "coefficients"  "model"

> s <-summary(fit)
> s

Call:
plr(formula = proficiency_mt ~ proficiency_lp + computers + classroom +
    employees + classroom_used, data = pol_dta_train)

Residuals:
   Min      1Q  Median      3Q     Max
-28.67   -4.16   -0.30    3.89   35.33

Coefficients:
                        Estimate Std. Error  z value  Pr(>|z|)
(center-intercept)     -8.9771931  1.9302980   -4.6507 3.384e-06 ***
center-proficiency_lp   1.0377575  0.0083011 125.0140 < 2.2e-16 ***
center-computers        0.0877895  0.0287689    3.0515  0.002287 **
center-classroom       -0.1530676  0.0983827   -1.5558  0.119803
center-employees       -0.0242880  0.0171906   -1.4129  0.157750
center-classroom_used   0.0654843  0.1037890    0.6309  0.528108
(radius-intercept)     24.1153891  1.3179920   18.2971 < 2.2e-16 ***
radius-proficiency_lp   0.7027862  0.0148911   47.1951 < 2.2e-16 ***
radius-computers        0.0272065  0.0047557    5.7209 1.114e-08 ***
radius-classroom        0.0094819  0.0102173    0.9280  0.353438
radius-employees        0.0019548  0.0032473    0.6020  0.547207
radius-classroom_used   0.0065859  0.0126222    0.5218  0.601851
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '
```

```
fitted_polygons <- fitted(fit, polygon = T, vertices = v)
head(fitted_polygons, 1)
[[1]]
          [,1]      [,2]
[1,] 318.6278 260.6925
[2,] 305.8107 285.8475
[3,] 285.8475 305.8107
[4,] 260.6925 318.6278
[5,] 232.8079 323.0443
[6,] 204.9233 318.6278
[7,] 179.7683 305.8107
[8,] 159.8051 285.8475
[9,] 146.9880 260.6925
[10,] 142.5715 232.8079
[11,] 146.9880 204.9233
[12,] 159.8051 179.7683
[13,] 179.7683 159.8051
[14,] 204.9233 146.9880
[15,] 232.8079 142.5715
[16,] 260.6925 146.9880
[17,] 285.8475 159.8051
[18,] 305.8107 179.7683
[19,] 318.6278 204.9233
[20,] 323.0443 232.8079


pplot(fitted_polygons) +  labs(x='Dimension 1', y='Dimension 2') + theme_bw()

rmsea(fitted_polygons, pol_dta_train$proficiency_mt)

[1] 3764.133

zeros <- matrix(0, nrow = nrow(center_test), ncol = ncol(center_test))
X1 <- cbind(1, center_test[, -2], zeros)
X2 <- cbind(zeros, 1, radius_test[, -2])
X <- rbind(as.matrix(X1), as.matrix(X2))
Y <- matrix(c(center_test[, 2], radius_test[, 2]))
beta <- (fit$coefficients)

Y_estimated <- X %*% beta
Y_center_estimated <- Y_estimated[1 : n_test]
Y_radius_estimated <- Y_estimated[(n_test + 1) : (2 * n_test)]


dta_estimated <- list()
dta_estimated[[1]] <- data.frame(y_estimated = Y_center_estimated)
dta_estimated[[2]] <- data.frame(y_estimated = Y_radius_estimated)
names(dta_estimated) <- c('center', 'radius')
class(dta_estimated) <- 'paggregated'

dta_estimated <- psymbolic(dta_estimated, v)

rmsea(dta_estimated$y_estimated, pol_dta_test$proficiency_mt)
4106.117

id_test[1 : 2]
dta_estimated$y_estimated[1 : 2] %>% pplot() + theme_bw()

pmean_id(dta_estimated$y_estimated[[1]])
pmean_id(dta_estimated$y_estimated[[2]])

sqrt(pmean_id(dta_estimated$y_estimated[[1]]))
sqrt(pmean_id(dta_estimated$y_estimated[[2]]))
```

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